



FAKE NEWS DETECTION

Using Machine Learning & Deep Learning Approaches

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Algorithms
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Embedding
Strategies

99.78%

Best Model
Accuracy

Introduction & Problem Statement



Why Fake News Detection Matters

- In today's digital era, information spreads instantly across social media platforms. Anyone can publish content without verification, bypassing traditional journalism gatekeeping.
- Fake news has significantly influenced major events like elections, COVID-19, and social movements. It can mislead people and negatively impact public opinion and decision-making.
- Misinformation can create panic, confusion, and harm public safety. It also reduces trust in digital platforms and reliable information sources.
- Manual fact-checking is slow and cannot keep up with the massive volume of online content generated every day. Millions of posts make human verification impractical.
- Therefore, there is a strong need for automated and intelligent systems. These systems can efficiently detect fake news and ensure reliable information dissemination.

+1M

News articles published daily

50%~

Social media shares are misinformation

99.78%

Our best model accuracy (RoBERTa, ISOT)

PROBLEM: Build an automated, scalable system to classify news as genuine or fake across diverse domains with minimal compute.

Classical ML Approaches

Core Idea:

Word2Vec learns vector representations of words such that words appearing in similar contexts have similar meanings.

Techniques:

- CBOW (Continuous Bag of Words): Predicts a word from its context
- Skip-Gram: Predicts context from a given word

Key Contributions:

- Introduced dense, low-dimensional word vectors (compared to sparse TF-IDF)
- Captures semantic relationships (e.g., king – man + woman $\approx$ queen)

In Fake News Detection:

- Helps capture subtle linguistic patterns and semantic meaning
- Better than TF-IDF for context-aware classification

Limitations:

- Context-independent (same word = same vector always)
- Cannot handle polysemy (e.g., "bank")

Word Embeddings & RNNs

Core Idea:

Global Vectors (GloVe) combines global word co-occurrence statistics with local context learning.

Key Features:

- Uses a co-occurrence matrix
- Learns embeddings by factorizing this matrix

Advantages:

- Captures both:
 - Local context (like Word2Vec)
 - Global statistics (overall corpus structure)

In Fake News Detection:

- Improves detection of semantic similarity across articles
- Works well for capturing topic-level relationships
-

Limitations:

- Still context-independent
- Requires large corpus for effective learning

Transformer Models

(Often used alongside Word2Vec in literature)

Core Idea:

Represents words as character n-grams instead of whole words.

Advantages:

- Handles:
 - Misspellings
 - Rare words
 - Morphological variations
 -

In Fake News Detection:

- Useful for noisy social media text
- Better performance on short and informal content

Datasets — Five Benchmark Corpora



Dataset	Size	Avg Length	Domain	Split
ISOT	45,000	~350-400 words	Political / General News	80-20
Fake & Real News	45,000	~350-400 words	Political / General News	80-20
LIAR	12,800	~20 words	Political Statements	80-20
GossipCop	22,000	~800 words	Celebrity / Entertainment	80-20
Twitter15/16	742 / 412	~15 words	Social Media Tweets	10-Fold CV

Long-Form

ISOT, F&R, GossipCop
Rich stylistic signal
High accuracy achievable

Short-Text

LIAR & Twitter
Sparse, ambiguous content
Challenging for all models

Diverse Domains

,Politics, Celebrity
Social Media
Cross-domain testing

Feature Extraction — Three Embedding Strategies



Strategy 1

TF-IDF Vectors

features $\rightarrow$ Naïve Bayes + SVM 50,000

Overview:

TF-IDF (Term Frequency–Inverse Document Frequency) is a statistical method used to convert text into numerical features based on word importance.

How it Works:

- Term Frequency (TF): Measures how often a word appears in a document
- Inverse Document Frequency (IDF): Reduces importance of common words
- Final score = $TF \times IDF$

Key Characteristics:

- Produces a sparse, high-dimensional matrix (e.g., 50,000 features)
- Each word is treated independently $\rightarrow$ no context awareness
- Does not capture word order or semantics

Advantages:

- Fast and computationally efficient
- Works well with traditional ML models like Naive Bayes and SVM
- Strong baseline for comparison

Limitations:

- Cannot understand meaning or relationships between words
- Fails on context-based interpretation (e.g., sarcasm, tone)

Strategy 2

GloVe / Word2Vec

dim dense vectors $\rightarrow$ LSTM + BiLSTM-100

Overview:

These are pre-trained word embeddings that convert words into dense vectors capturing semantic meaning.

Word2Vec:

- Uses CBOW / Skip-Gram models
- Learns word relationships from local context

GloVe:

- Uses global co-occurrence statistics
- Trained on large corpora like Wikipedia (6B tokens)

Key Characteristics:

- Produces dense, low-dimensional vectors (e.g., 100–300 dimensions)
- Captures semantic similarity (e.g., “king” $\approx$ “queen”)
- Static embeddings $\rightarrow$ same vector for a word everywhere

Advantages:

- Better semantic understanding than TF-IDF
- Works well for capturing context flow in sentences
- Improves performance in fake news detection tasks

Limitations:

- Cannot handle polysemy (same word, different meanings)
- Context is limited compared to transformers

Strategy 3

BERT Contextual

dim [CLS] embeddings $\rightarrow$ RoBERTa + ALBERT-768

Overview:

BERT (Bidirectional Encoder Representations from Transformers) generates context-aware embeddings using transformer architecture.

How it Works:

- Uses self-attention mechanism to understand relationships between all words in a sentence
- Processes text bidirectionally (left + right context)

Key Characteristics:

- Produces dynamic embeddings $\rightarrow$ same word gets different vectors based on context
- Uses special tokens like [CLS] for classification tasks
- Embedding size typically 768 dimensions

Tokenization:

- Uses WordPiece / SentencePiece, handling unknown and rare words effectively

Advantages:

- Captures deep semantic + syntactic meaning
- Handles ambiguity and context very well
- Achieves state-of-the-art performance on NLP tasks

Limitations:

- Computationally expensive
- Requires GPU/TPU for efficient training

Proposed Methodology — Six Classification Algorithms



Classical ML

Naïve Bayes

- Uses probability to classify news using TF-IDF features (~50K features)
- Assumes words are independent of each other
- Very fast model with low training time
- Gives decent accuracy (~70–80%) as a baseline

Linear SVM

- Finds the best boundary to separate real and fake news
- Works well with high-dimensional data (TF-IDF ~50K features)
- More accurate than Naïve Bayes (~80–85%)
- Fast and efficient for large datasets

Deep Learning (RNN)

LSTM

- Neural network that processes text sequentially
- Uses embeddings (100–300 dimensions)
- Remembers important past information in sentences
- Achieves better accuracy (~85–90%)

BiLSTM

- Reads text in both forward and backward directions
- Uses 64–128 units for better context understanding
- Captures full sentence meaning more effectively
- Improves accuracy (~88–92%) over LSTM

Transformer-Based

RoBERTa

- Transformer-based model with ~12 layers and 768-dim embeddings
- Pre-trained on large datasets (billions of words)
- Understands deep context and semantics
- High accuracy (~92–96%) in fake news detection

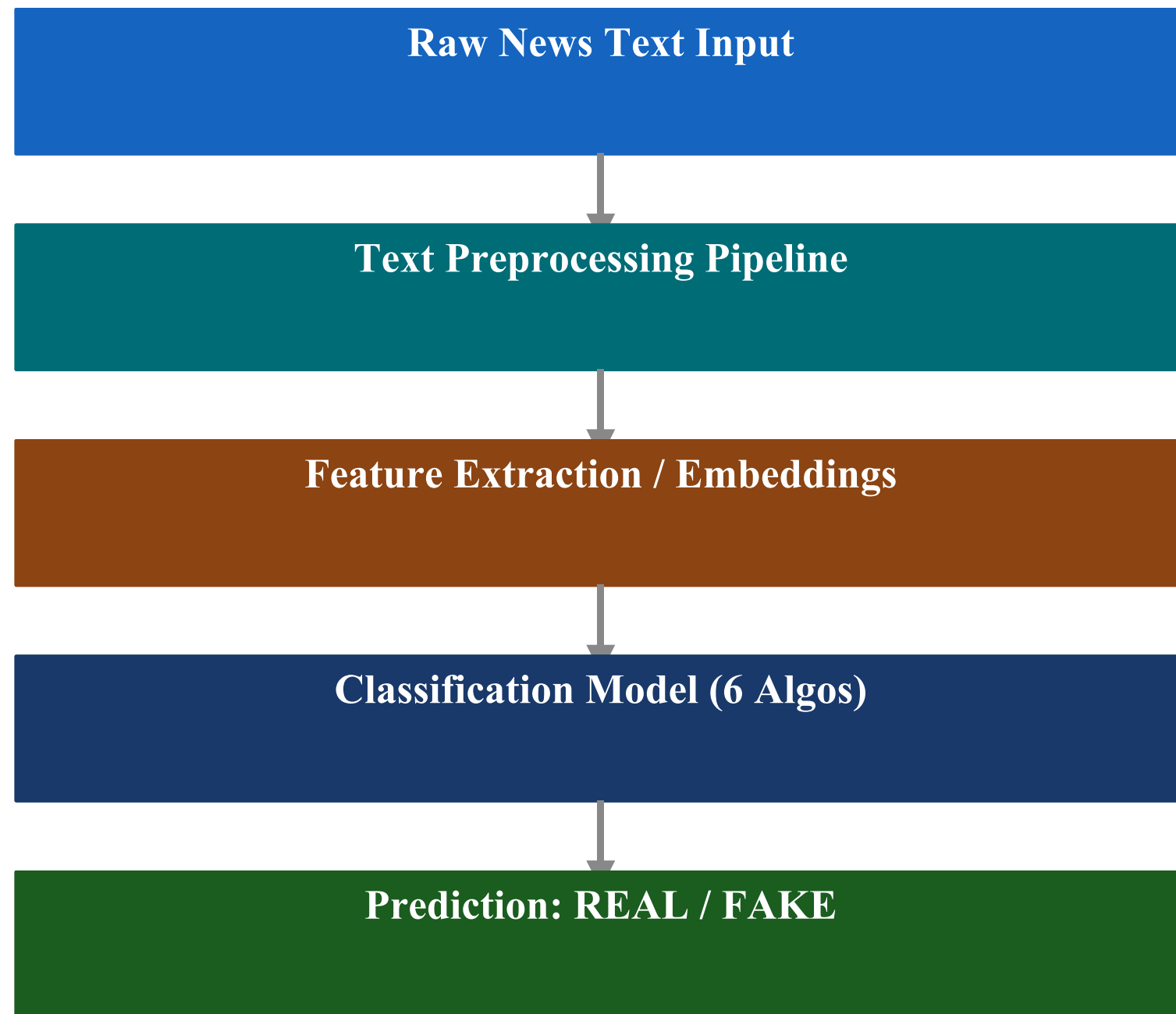
ALBERT

- Lightweight version of BERT with ~89% fewer parameters
- Faster and memory-efficient than RoBERTa
- Uses shared parameters for efficiency
- Accuracy slightly lower but still strong (~90–94%)

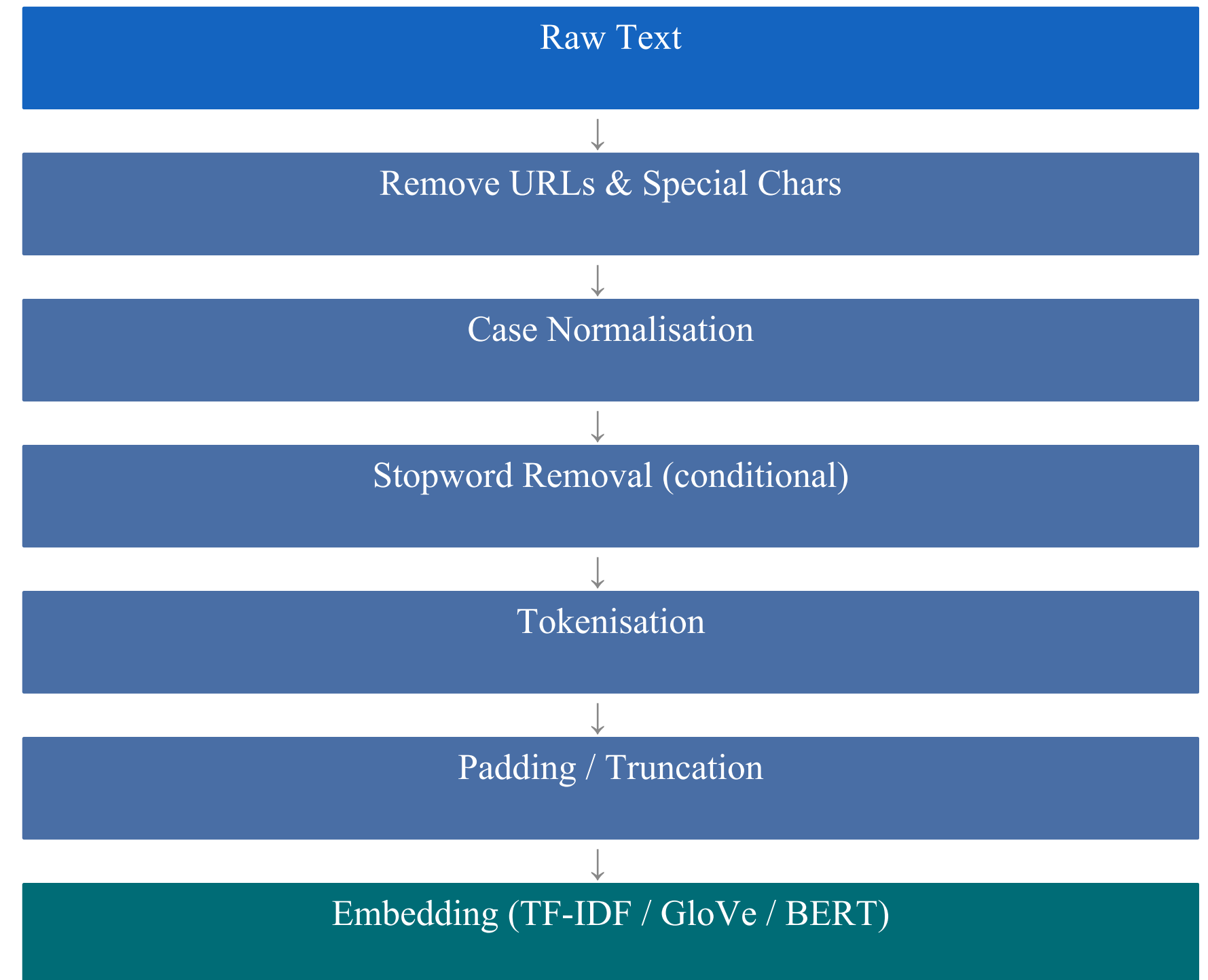
System Diagram / Proposed Algorithm



Overall System Architecture



Preprocessing Pipeline Flowchart



Implementation



Libraries & Frameworks Used

Category	Tool / Library	Purpose
ML Models	scikit-learn	Naïve Bayes, LinearSVC, TF-IDF vectorization
Deep Learning	TensorFlow 2.x / Keras	LSTM, BiLSTM training & GPU acceleration
Transformers	Hugging Face Transformers	RoBERTa, ALBERT fine-tuning
Embeddings	GloVe (6B), Gensim	Word2Vec Skip-Gram training
Data Processing	pandas, numpy	Dataset loading, preprocessing, train/test split
Evaluation	sklearn.metrics	Accuracy, F1, confusion matrix, ROC-AUC
Visualization	matplotlib, seaborn	ROC curves, confusion matrices, comparison plots

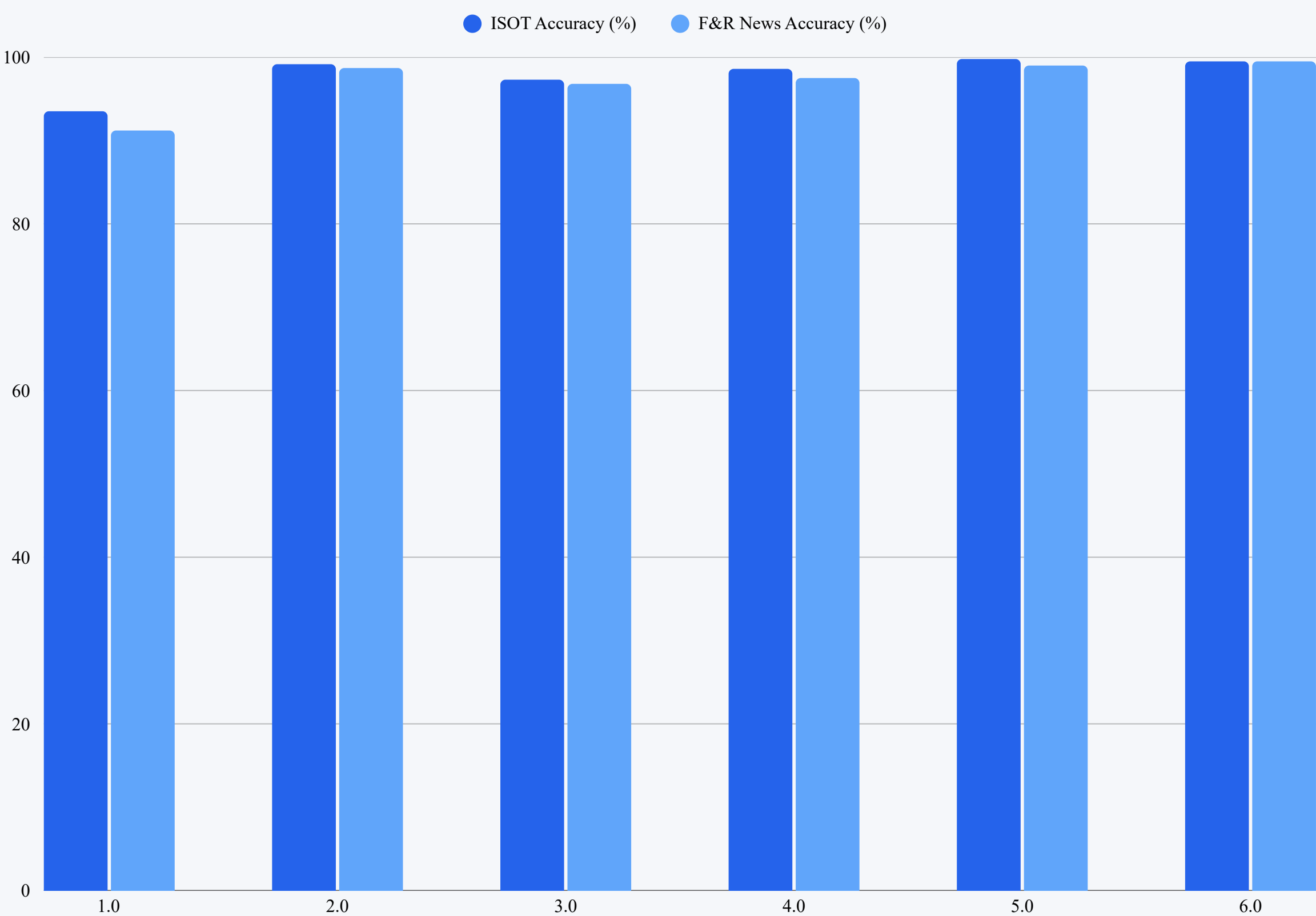
Hardware Configuration

- CPU: Intel Core i7 (or equivalent) with 16GB RAM
- GPU: NVIDIA Tesla T4 (Colab) / RTX 3060 for transformer fine-tuning
- Storage: 50GB+ for datasets and model checkpoints
- Environment: Python 3.9+, Google Colab Pro / local Conda environment

Results, Comparison & Analysis



Long-Form Datasets (ISOT & Fake & Real News) — Accuracy %



99.78%

RoBERTa on ISOT
(Best overall accuracy)

99.16%

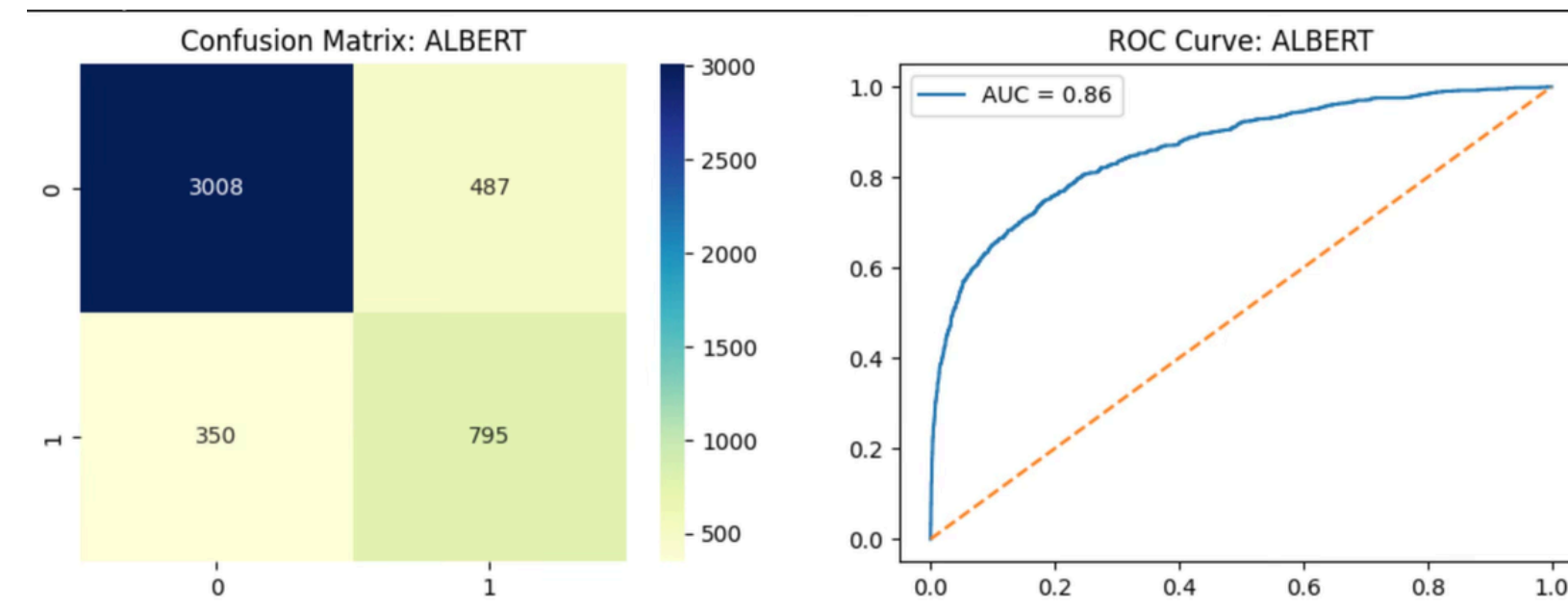
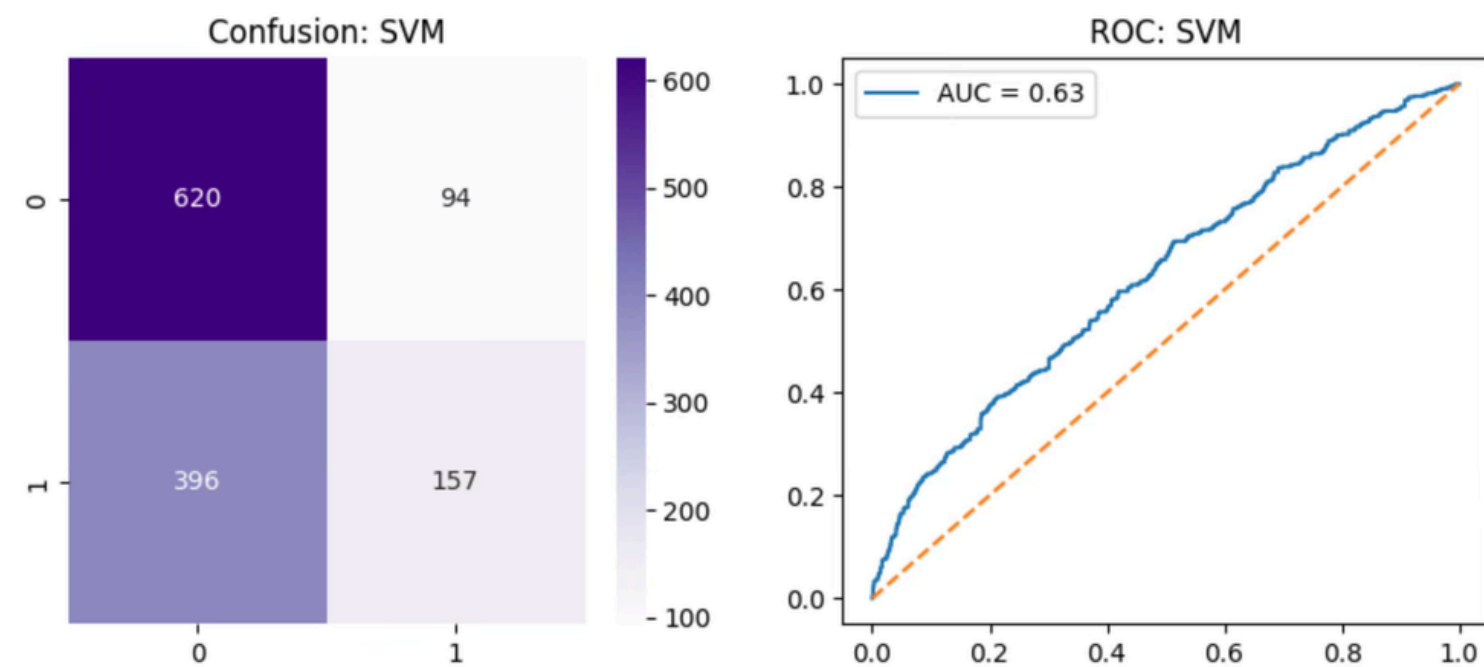
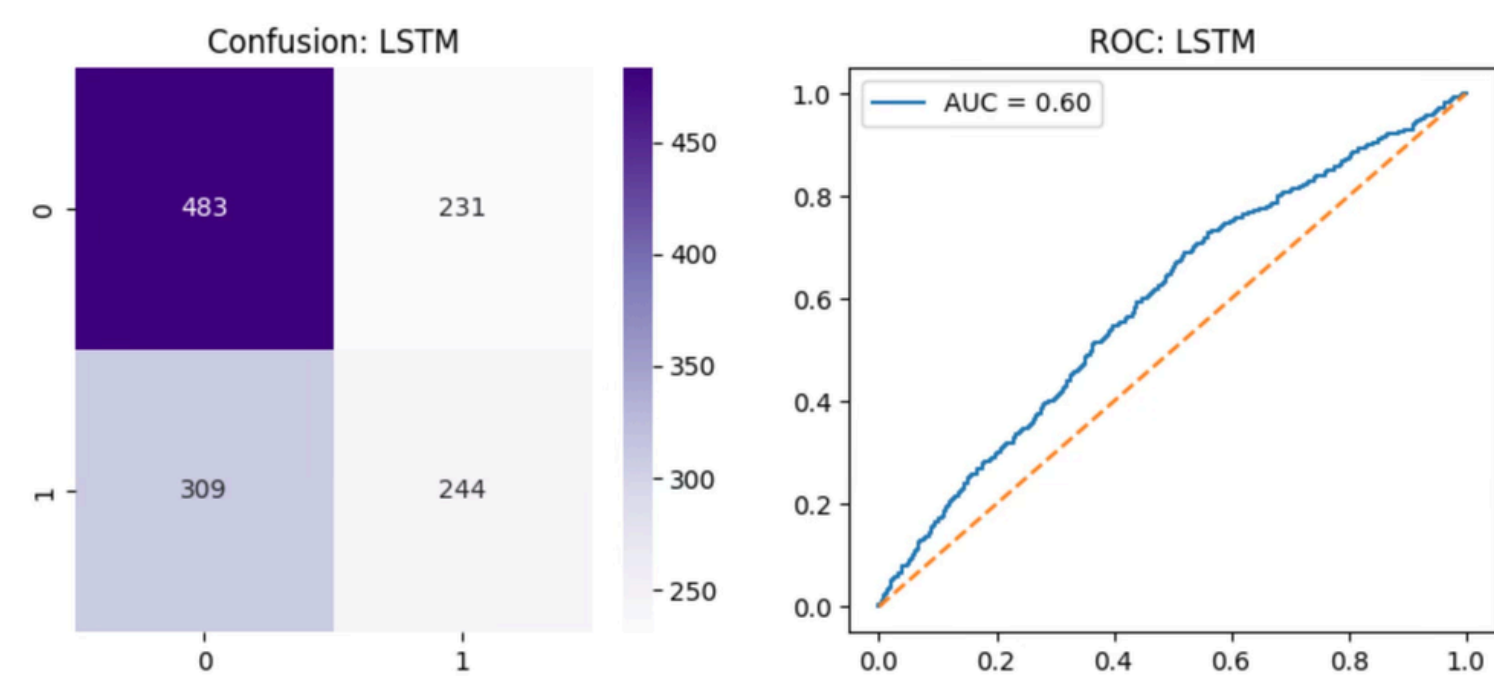
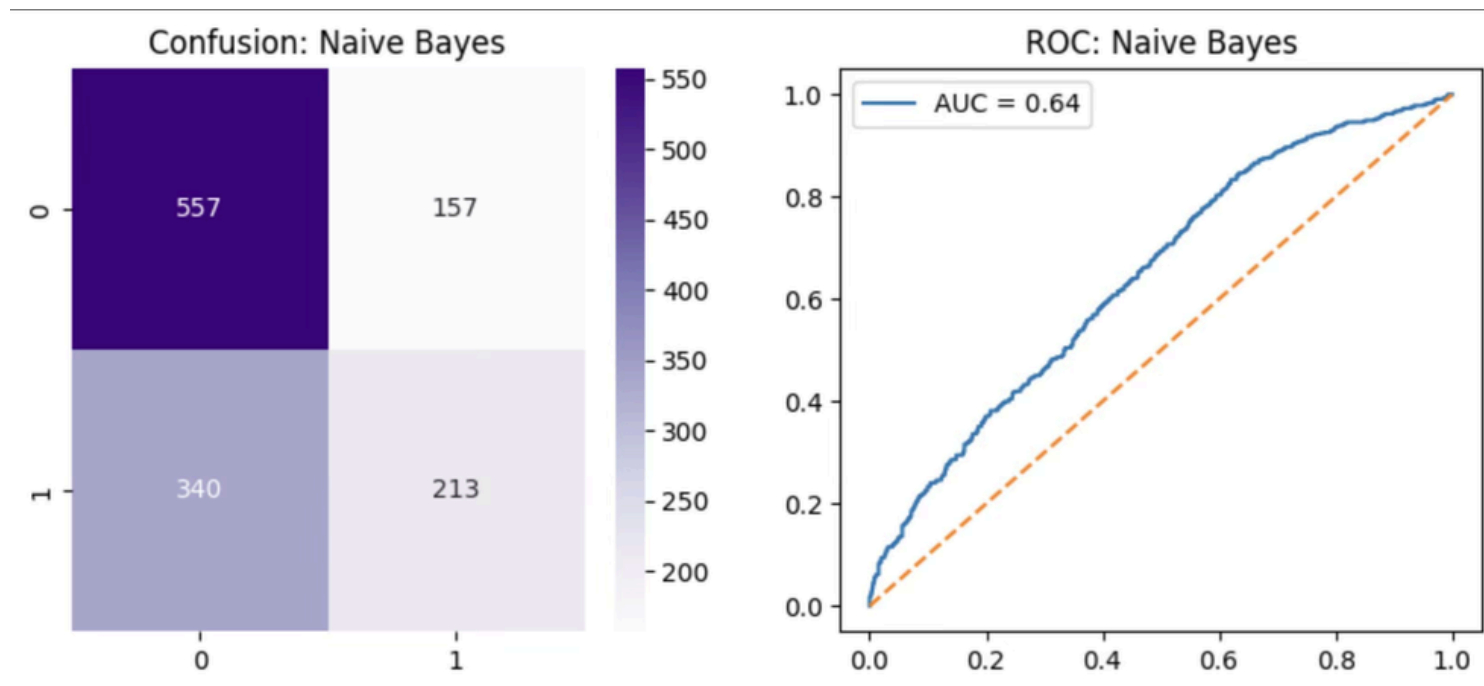
SVM on ISOT
(faster than RoBERTa ×50)

93.51%

Naïve Bayes on ISOT
(Strong classical baseline)

Short-Form Datasets — F1 Score

Model	LIAR	GossipCop
SVM	62.9%	89.0%
BiLSTM	64.1%	91.2%
RoBERTa	68.5%	96.1%



Results — Key Finding: Embedding Quality > Classifier Complexity



F1 Score Gain on Twitter15 — TF-IDF → BERT Contextual Embeddings | Same BiLSTM 10.3%+ Architecture

LIAR Challenge

Max F1 only 68.5% — short political statements (avg 20 words) lack sufficient discriminative textual signal.

GossipCop Success

RoBERTa achieves 96.1% F1 — longer celebrity articles enable strong cross-domain generalization.

Twitter Limitation

Content-only models peak at 79.4% vs. 82.7% F1 for graph-propagation methods — propagation structure matters.

ROC-AUC Scores by Model & Dataset

Model	ISOT	LIAR
Naïve Bayes	0.98	0.64
SVM	1.00	0.63
LSTM	1.00	0.60
GloVe LSTM	0.84	—
W2V LSTM	0.66	—
RoBERTa	0.90	~0.75

This gain (+10.3%) exceeded every improvement from changing model architecture — proving that investing in better embeddings is more cost-effective than building more complex classifiers.

Conclusion & Future Work



RoBERTa Dominates

99.78%

on ISOT dataset

-Best for high-resource production deployment. Near
.perfect on long-form news

SVM = Best Efficiency

99.16%

at 50× lower compute

Ideal for edge / real-time deployment with
.constrained resources

Embedding > Architecture

10.3%+

F1 from TF-IDF → BERT

Invest in better representations first before complex
.classifiers

DEPLOYMENT RECOMMENDATIONS

★ High Accuracy Required

*RoBERTa / ALBERT + BERT
embeddings*

⚡ Compute-Constrained

Linear SVM + TF-IDF (50K features)

🔗 Short Social Media Text

BiLSTM + BERT + propagation trees

✓ Multilingual / Scalable

*ALBERT (89% fewer params, same
perf.)*

Future Scope: Multimodal detection (text + image) · Graph propagation trees for Twitter · Multilingual BERT fine-tuning · Larger transformer variants

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Thank You

Questions & Discussion